

CHAPTER 2.1 - THE GRID DOESN'T NEED A COMMANDER

I. The same fight, twice

I.1 1993

My working life began inside the Dutch telephone company KPN Telecom. It has just been unbundled from the post office in 1989 but it still ran the way it had run for decades. The architecture had a name engineers used without irony – the *intelligent network* – and inside that name there lived a complete worldview. Intelligence sat in the central exchanges. The instrument on your desk was, by deliberate design, as dumb as a brick: a microphone, a speaker, a dial pulse, nothing more. The job of the network was to set up a circuit between two endpoints, hold it open for the duration of the call, and tear it down at the end. Quality, billing, features, security, lawful intercept – all delivered from the centre.

It is easy, thirty years later, to be patronising about this. Don't. The intelligent network was not a relic. It was a deliberate, sophisticated solution to a deliberate, sophisticated problem: *how do you guarantee that any phone in the country can reach any other, in seconds, with reliable voice quality, while billing for the privilege?* The solution was elegant: make the network able to take charge of every call while making the device at each end as simple as possible. Centralise everything that needs to be centralised, and centralise quite a lot more besides.

The engineers who built that network were not fools. There was a coherent technical caste behind it, known in the trade as the *Bellheads* after AT&T's [Bell Labs](#), and KPN had its own equivalent. Bellheads possessed a coherent worldview backed by a coherent technology backed by a coherent business model. The worldview was that telephony was hard, the technology was the intelligent network, and the business model was billing for minutes. These three things had been mutually reinforcing for almost a century, which is the kind of streak that tends to make an industry -- whether it's Telecom or the electricity grid -- feel that the next century will look more or less the same.

I.2 The fight nobody at the top wanted to have

For me, fresh out of my student days, the Bellheads and their architecture seemed decidedly odd. I was friends with a guy whose startup company later coined the term WiFi. His friend, started the first Dutch internet service provider, [XS4ALL](#) (Access for All). I worked a student job at the computing centre of the [Technical University of Twente](#) where I got to chat with people all over the world using this new-fangled thing called the Internet. We all thought the future belonged to TCP/IP: the internet protocol. In telecom, such people were later called Netheads.

With TCP/IP you created a network built on packet switching instead of circuit switching. The network provided no setup, no teardown, no guaranteed circuit – each packet found its own way through whatever routers happened to be available. It was the inverse of the

telephone network. Dumb in the middle, smart at the edges. Devices at the edges decided what to send and how fast. The network's job was simply to forward whatever it got, and to warn when it couldn't.

When I started working in telecom, TCP/IP was fringe and most managers and engineers fantasized about a future which held a more sophisticated version of the present. The favoured candidate was [ATM – Asynchronous Transfer Mode](#) – a connection-oriented packet protocol with quality-of-service guarantees and per-flow accounting. ATM was the Bellhead vision of how packet switching could be done *properly*: keep the network in charge, give it knobs to turn, preserve the operator's role as the responsible adult. The standards body for ATM was the [ITU](#), the formal United Nations agency in which national operators had voting power. The standards body for IP was the IETF and it selected its protocols by what it cheerfully called [rough consensus and running code](#). One body wore suits. The other body wore T-shirts. Both bodies thought the future depended on the other one going away.

In 1997 a research engineer at AT&T named David Isenberg published an essay called [The Rise of the Stupid Network](#). The argument was as simple as it was rude to the Bellheads: the intelligent network was being eaten by a stupid one, because intelligence at the edge scales and intelligence in the centre does not. Isenberg used the word *stupid* on purpose, as a way of throwing it back at the people who had been using it to dismiss IP. AT&T did not enjoy the essay. [Isenberg left soon after](#). [Tom Evslin](#), an AT&T executive at the time, later recalled that reading it was like getting a glass of cold water in the face (which he -- in retrospect -- meant as a compliment).

[1.3 How it actually ended](#)

KPN did not become an internet company because they read Isenberg and relished the cold water in the face, nor because a few people like me evangelised TCP/IP. KPN became an internet company because, starting in the late 1990s, [cable TV operators](#) began offering all the services KPN was offering. They achieved this by running TCP/IP over the coaxial cable that had previously only provided television. Their all-in-one offering was much cheaper, and customers started to desert KPN in droves.

The architectural lesson is unromantic: *the incumbents did not reform by themselves*. The disruption arrived from outside the regulatory envelope built around the incumbents – cable was a different industry under a different regulator – and the customers walked. The Bellheads were structurally incapable of initiating this move towards the internet architecture. Their institutional logic, their technical training, their billing systems, their relationships with national regulators were all organised around the worldview that the new architecture made obsolete. They were not bad people. They were the wrong people.

[1.4 The same shape, in electricity](#)

After all that – and after leading part of a project that switched everything within KPN onto the Internet Protocol – I felt I had reached an end. I had gone from one of the few lonely idiots promoting TCP/IP to nobody disagreeing any more. I decided I needed a sabbatical. During that sabbatical I discovered fossil fuels were losing due to the ongoing technological revolutions in solar, wind, batteries and electric vehicles, and it felt vaguely

similar and at least as important. I decided this was my chance to start a brand new career. Now it's 2026, but in many ways it feels like 1993 all over again.

The grid now, it seems to me, is in roughly the position telecom was in around 1993. It has gone through one architectural reform – the unbundling that began with the [EU's First Electricity Directive in 1996](#) and was implemented in the Netherlands as the [Elektriciteitswet of 1998](#) – and that reform forced the invention of wholesale markets coordinating energy supply. Now it's possible for anybody to provide energy, just like the internet made it possible for anybody to start a blog and provide content.

But in terms of energy transport, this is still a network that ignores the intelligence at the edges and likes consumer equipment that is as dumb as a rock. And even though the electricity grid is a lot dumber in terms of routing and monitoring than the Internet, the intelligence that remains is in the centre.

The grid operators organise everything centrally, even though there are now massive amounts of intelligence and flexibility at the edges. All this intelligence and storage is largely unused, because it does not fit the centralised paradigm. There is a market for content, but not for transportation.

An alternative is possible. The grid could use all the intelligence and storage at the edges to use the grid far more optimally. It could use that same intelligence and storage to keep power flows and voltage stable. It could use simple protocols, similar to the internet protocol, to route energy in such a way that grid nodes that are in danger of being overloaded get bypassed using other nodes that are not. For somebody with an internet background, it all seems so logical, so efficient, so safe, so scalable, so resilient. And it has been proven in countless small pilots. We know it works. Yet it does not progress any further than that, because of the Bellheads of the grid.

The Bellheads of the grid are not bad people. They built a system that has kept the lights on in our cities for over a century. In many parts of the world it's so reliable that almost nobody under fifty has had to think about how it works. The institutional logic is built around protecting that. When the Bellheads of the grid look at distributed price coordination – markets at the substation, markets in the street, markets in the wall box – what they see is a million failure points where today there is one reliable operator. They are wrong about that, but they are not wrong for cynical reasons. They are wrong because their training, their tools, and their working lives have been organised around an architecture that is reaching the end of its useful life.

The Netheads of the grid exist. The [DLMP](#) theorists, the [PowerMatcher](#) builders, the [Olympic Peninsula](#) and [DSO+T](#) operators, the auction designers we shall meet in this chapter. For two decades they have been building, demonstrating, and being told to come back later.

As in 1993, the disruption that will end this fight is unlikely to come from inside. But as in 1993, it is already arriving from outside. Rooftop solar is profitable regardless of whether the regulator wants it. As household batteries become cheaper, they become profitable whether the grid operator wants it or not. Electric cars and electric trucks turn out to be

the largest source of flexibility ever invented, and can either become the electricity grid's biggest nightmare or its biggest boon. There are countless companies packaging these new technologies in ways similar to the cable companies in the telecom revolution. They will keep forcing the grid operator's hand until it learns to embrace localized markets.

II. Why markets beat commanders

II.1 The knowledge problem

The solution we are proposing did not arrive with the internet protocol. It is far more generic and far older. Nature has been using this approach for billions of years, but let me skip to 1945 and to [Friedrich Hayek](#). He was arguing against the Bellheads of his day, who believed a sufficiently clever planner could allocate everything from the centre. In *The Road to Serfdom*, Hayek had argued that it was an easy slide from a centrally planned economy to an authoritarian one. Now he turned from ideology to practicality. He argued that a central planner would never be as optimal as a decentralized market, for a simple and generic reason: the relevant knowledge required to make optimal decisions is always dispersed. *You* know when you feel like getting up in the morning. *You* know when your partner would like a cup of tea. *You* know when you have to take your daughter to piano lessons. It is impossible for a central planner to know all that in as timely and detailed a manner as you.

II.2 The zero-duality gap

But even in the theoretical situation that a central planner would know everything a local market would know, there would still be no advantage to central planning over decentralised planning. This is something many people intuitively do not expect. Many of us come from a tradition in which we expect a king, president, hero, or enlightened despot would theoretically be able to take the best decisions. It can, however, be mathematically proven that this is not the case. To return to the electricity grid: it turns out, [under conditions Boyd and Vandenberghe set out clearly in their standard textbook](#), that exactly the same solution as reached by a central planner can be reached by a multi-level market of bidders adjusting prices and quantities under transparent constraints. This mathematical proof is often referred to as the *zero-duality gap*. In simple language: even in the unrealistic situation where a central planner has perfect knowledge, they still cannot improve on a well-structured multi-level market.

On top of this, multi-level markets break up the optimisation problem into smaller parts, and this has two advantages. First, if some local market fails, the markets above, below, or next to it do not. The failure is localised. When a central planner fails, that failure takes the whole system down. Multi-level markets are therefore more resilient. Secondly, finding the optimal solution for a small market is much easier, and therefore small markets can adjust prices faster. This in turn means they can counter shocks faster – which adds to their already superior resiliency. Add to this Hayek's argument that local markets have more knowledge, and it is clear that well designed local markets are superior to centralised top-down planning.

All of this is, of course, also in line with Simon's *near-decomposability* and the term *holons* we chose for it (following Koestler) that we presented in the previous chapter. Koestler, Simon, Boyd and Vandenberghe, Hayek – they are all saying the same thing. In the field of governance, giants like [Ostrom](#) pile on. And in biology, ecology, trade and organisational science, holons are by now usually the standard, even though the terms used differ from domain to domain.

III. Why the current grid market is incomplete

For most of the twentieth century the grid was not a system that needed coordination between actors, because there was, in any given region, one actor. A vertically integrated utility owned the generators, the transmission lines, the distribution wires and the meter on the wall. Generation followed forecast demand by direct command from a control room. Costs were aggregated and recovered through a regulated tariff. Electricity was sold by the kilowatt-hour at a price that bore no detectable relation to the marginal cost of supplying it at any given moment. There was no market and no marginal price. There was a balance sheet and a regulator.

That changed with unbundling. The [EU First Electricity Directive in 1996](#) required member states to separate generation from transmission ownership; the [Dutch Elektriciteitswet of 1998](#) implemented the directive at home. Similar unbundling had already begun in the United States with [PJM's wholesale market in 1997](#) and in the Nordics with [Nord Pool going Nordic in 1996](#). After unbundling, the same generators that had previously been dispatched by a control-room operator were now owned by separate companies. They could no longer be told what to do. They had to be paid. The wholesale market was invented to do that paying, in real time, for the marginal cost of meeting demand minute by minute.

The result is what people now call the wholesale stack. It is a frankly Byzantine set of markets, but let us pick out the four most important ones. *Day-ahead*: every producer and large consumer submits bids by noon for each hour of tomorrow. The exchange clears. Generators are dispatched. *Intraday*: as the day approaches and forecasts sharpen, market participants buy and sell residuals. *Balancing*: in the operating hour itself, the transmission system operator procures upward and downward reserves to match instantaneous supply with instantaneous demand. *Frequency control*: faster still, automatic, primary and secondary reserve, operating on physics rather than markets. Each is a centralised market over the entire bidding zone, with prices that result from auctions. Auctions are cleared by one commander for the entire bidding zone, which is often a country.

The current grid uses the wholesale stack of markets to coordinate supply and demand over time. It's not a perfect system by any stretch of the imagination. It's too complicated, opaque, disintegrated and slow for that. But it's something.

What's entirely missing is the spatial and temporal coordination of grid capacity. Every home and every business sees the same transport cost, irrespective of where and when grid congestion occurs. As a result the grid is used far less efficiently than energy supply. This is what is causing the ballooning grid costs in the Netherlands and what caused the recent blackout in Spain.

IV. What the missing market looks like

IV.1 PowerMatcher (2005)

The alternative is not a hypothetical. In 2005 [Koen Kok at TNO, with collaborators in Eindhoven and Alliander, published a paper](#) describing how to implement the missing market under the name PowerMatcher. All flexible devices bid directly. Local prices are cleared by an auctioneer. This can be done multiple times per second, if need be. Each device takes one constraint into account: the transformer's capacity limit. A heat pump knows what indoor temperature its owner is willing to tolerate, and moves its energy demand in time so the energy is cheapest and the transformer is not overloaded. An EV acts similarly, taking into account when it has to leave fully charged. A battery knows what its arbitrage opportunity is. Each translates that private knowledge into a bid curve. The auctioneer adds the curves up. The price falls out.

Kok later described the architecture, in his [doctoral thesis](#) and again in [a 2016 IEEE Power & Energy Magazine paper with Steven Widergren](#), as a *Society of Devices*: heat pumps, EV chargers, batteries and white goods as economic agents with bid curves, negotiating continuously. The grid operator is no longer the conductor. It is the referee. It announces the constraint. The market works out who pays what to use the available capacity.

Sidebar A. A worked auction. Imagine a substation transformer with 100 kW of spare capacity for the next fifteen minutes. Behind it sit five EVs, each willing to pay something to charge, but each with a different urgency. EV 1 has to leave at seven and is empty: it bids 30 cents per kWh up to 11 kW. EV 2 is parked all night and will accept anything below 8 cents up to 11 kW. EVs 3, 4 and 5 sit in between with various combinations. The auctioneer stacks the bid curves from highest to lowest and walks down them until the cumulative demand hits the 100 kW transformer limit. Wherever that crossing happens, the price clears. EVs above the line pay the clearing price and charge; EVs below the line wait. Nobody negotiated anything. Nobody told a charger what to do. The transformer was respected. The most urgent EV charged first because its owner was willing to pay for it. Repeat in fifteen minutes with the new bidders, the new state of charge, the new transformer capacity. The grid operator, who built the substation, now collects a congestion rent from the difference between the clearing price and what the system above is paying – money that pays to upgrade the substation faster than a flat tariff ever would. No commander. One number. Cleared.

IV.2 Layered Decentralised Optimisation & DLMP (2016–2018)

Over the next decade the academic literature put new names on what Kok had built. [Kristov, De Martini and Taft, in 2016](#), called it *Layered Decentralised Optimisation*: each layer of the grid – building, street, substation, bidding zone, inter-zone – clears its own market, presents its residual to the layer above, and accepts the layer above's price as its boundary condition. Below the layer, opaque. Above, opaque. Between layers, one number. [Bai and colleagues, in 2018](#), worked out the formal mathematics under the name *Distribution Locational Marginal Pricing*: every bus in a distribution network gets its own

price, decomposed cleanly into five components – energy, losses, congestion, voltage, stability. In effect, every building gets its own bidding zone.

Muto and colleagues, also in 2018, answered the deployability question. The auction converges robustly even with devices joining and leaving mid-clearing – the *plug-and-play* property the grid actually needs, because of course not every appliance will participate, and of course participation will rise and fall through the day. The auction simply clears with whoever happens to be there.

V. Practical Examples

V.1 Olympic Peninsula (2007)

Hammerstrom and colleagues, at the Pacific Northwest National Laboratory, ran the first serious field trial. 112 households on the [Olympic Peninsula](#) of Washington State, each equipped with smart thermostats, water heaters and dryers; an auction clearing every five minutes; a transformer-capacity constraint the auction was tasked with respecting. The Result: peak load down by something close to 15%, no loss of comfort that the participants reported, and the auction never failed to clear.

V.2 GridWise and USEF (2010s)

PNNL ran a sequence of GridWise testbeds across the 2010s, the [American Energy Innovation Council compiled the lessons in 2019](#), and European replication ran through the [USEF programme](#) and [Universal Smart Energy Framework pilots](#) in Belgium, the Netherlands, and Germany.

V.3 DSO+T (2022)

In 2022 [Reeve and colleagues at PNNL](#) ran DSO+T, the largest co-simulation of transactive energy to date. They coupled wholesale and retail markets through transactive prices and ran the result across the full ERCOT footprint. The verdict: substantial savings, no instability, no comfort loss. Whatever the grid engineers were worried about, the simulation did not find it.

VI. So Why Isn't the Grid Holarchic

VI.1 Fairness – the strong but wrong objection

Locational and temporal pricing for transport could make people pay more or less, depending on where they live or what their consumption pattern is. Some areas and poor people who cannot buy flexibility might suffer. [Cahana, Fabra and Wang on the Spanish case](#) and Galeano-Suarez document the concern and it was also *the* concern the minister in the Netherlands mentioned when rejecting the idea recently. It is the strongest counter-argument the Bellheads of the grid can make.

There are also multiple counter-arguments and solutions. First of all: when this makes grid costs balloon, everyone has to pay more, including those supposedly helped by foregoing locational marginal prices. Secondly: the objection assumes prices are *transferred directly* to consumers' bills without compensation. But that's not a given at all. A holon is the

natural unit of solidarity. Inside a holon – a building, a street, a neighbourhood – members can choose to pool. Outside the holon, between holons and between regions, the price is a signal, not a tax. The signal between the neighbourhood and the substation is sharp. The settlement within the neighbourhood is human. This is the [Kaplow and Shavell \(1994\)](#) result on efficiency-equity separation: distort prices to deliver efficiency, redistribute through tax-and-transfer to deliver fairness, do not blunt the prices in the name of fairness because that costs you both. [Cramton and Stoft \(2006\)](#) spell out how to do it for electricity: split the bill into a fixed component and a variable component. If anybody's variable bill goes up unfairly, lower their fixed component until the total feels right. Setting the two knobs is politically fiddly. It is not, however, harder than rebuilding the grid in the absence of price signals – and that, increasingly, is the alternative.

VI.2 The institutional refusal

So now that we know the fairness argument has an answer, we can conclude the real obstical is probably different. I think it's institutional.

[Directive \(EU\) 2024/1711](#) – the European market design reform that came out of the 2022 energy crisis – explicitly required member states to allow for differentiated network tariffs that reflect locational scarcity. Compliance has been uneven. The [Commission's February 2025 infringement package](#) names the Netherlands among the laggards.

The Netherlands has been here before. [Motie Kroger of February 2024](#) asked the minister to investigate whether locational tariffs could ease the country's now-famous grid congestion. The [minister's reply in November 2024](#) was, in essence: we have looked at it and we will not do it. The reply rested in part on a 2019 [report by Ecorys and D-Cision, commissioned by the regulator ACM](#) that concluded locational tariffs were not necessary in the Netherlands. The report was written, as it happens, in 2019, before the congestion the Netherlands has since acquired and which has now closed parts of the country to new industrial connections.

The Netherlands more or less invented modern decentralised governance, through the Thorbecke reforms we already met. That it's now so strongly resisting nested pricing is ironic.

The Bellheads of the grid are not always wrong. Quality of service, security, settlement integrity – the things the intelligent network did well – are real concerns, and they are *more* real for the grid than for the telephone network. A misbehaving Netflix stream cannot crash the internet. A misbehaving battery, badly programmed, can crash a feeder. But you should treat such concerns as engineering with answers like certified inverter EMS software and well-defined fallback behaviour. It's not ok morph it into a refusal to adopt better architecture.

But I'm afraid the institutional pattern is what it is. The grid will not reform from inside. I hope nobody takes that verdict personally. I know from personal experience that neither the Bellheads of telecoms, nor the Bellheads of the grid are bad people. But their training, their tools, and their working lives have been organised around an architecture they cannot themselves replace. They are the wrong people for the job that comes next.

VII. The cable companies of the energy transition

As in 1993, the disruption that will end this fight is not coming from inside. As in 1993, it is already arriving from outside.

Rooftop solar is profitable regardless of whether the regulator wants it to be. [Household batteries](#) are profitable whether the grid operator wants it or not, and getting cheaper every year. Electric cars and electric trucks turn out to be the largest source of flexibility ever invented, and can either become the grid's biggest nightmare or its biggest boon depending on whether the grid learns to bid for that flexibility or refuses to. Grid-forming inverters – the subject of the chapter after next – will hold a feeder's voltage and frequency up locally, without asking permission, in places the central grid is too slow or too saturated to reach.

Each of these is a coax cable. Infrastructure outside the regulatory envelope built around the incumbent architecture. They do not require permission. They are getting cheaper. And there are countless companies packaging them into services – aggregators, virtual power plant operators, smart-charging providers, energy communities – in roughly the way the cable companies packaged TCP/IP over coax in the late 1990s. They will keep arriving. The grid will eventually bid for what they offer, or be bypassed by them.

VIII. The grid doesn't need a commander

The architecture is settled. The mathematics is settled. The pilots are settled. The chapter title is therefore a flat statement of fact rather than an aspiration.

KPN did not become an internet company because its engineers changed their minds. It became an internet company because cable companies came in over coax that was outside its regulatory envelope, customers walked, and Skype turned voice into a free feature of the new architecture. The Bellheads were structurally incapable of building the network that beat them. They were displaced by infrastructure their regulator did not control. The grid is now where telecoms was around 1993. The question is how we create something that puts as much pressure on the grid operators as the cable companies did on the telecom operators.